

Freight Rate Prediction

A concise summary of the data preparation, validation design, model selection, and final outputs.

\$127.09	\$646.94	0.8209
Holdout MAE	Holdout RMSE	Holdout R ²

1. Objective and data

The objective is to estimate **posted_rate** for every load in the unlabeled validation set. The training data contains 48,000 labeled loads from January through October 2025; the validation data contains 12,000 later loads from November and December. Each load includes locations, coordinates, distance, equipment, weight, date, market index, and quote signal.

2. Data preparation

The raw datasets were preserved and cleaned copies were used for analysis and modeling. Negative weights were treated as invalid, missing-value indicators were retained, and missing or invalid weights were filled with equipment-specific means calculated from valid training weights. Missing market-index values were filled using the training median. The same rules were applied to training and validation data.

The date was converted into month, day, weekday, and elapsed-day features. Geographic features included great-circle distance and the difference between trucking distance and straight-line distance. Pickup, delivery, equipment, and route were retained as categorical information for CatBoost.

3. Exploratory findings

- Distance is the strongest single numeric signal; its correlation with posted rate is approximately 0.91.
- Average rates differ by equipment: Reefer is highest, followed by Flatbed and Dry Van.
- The target is right-skewed, with a median near \$2,031 and a small number of very high-rate loads.
- Geographic features improved the time-holdout model from MAE \$216.66 to \$203.59 in the controlled comparison.

Validation, model selection, and outputs

The final workflow was evaluated as a future-prediction task.

4. Data split and validation

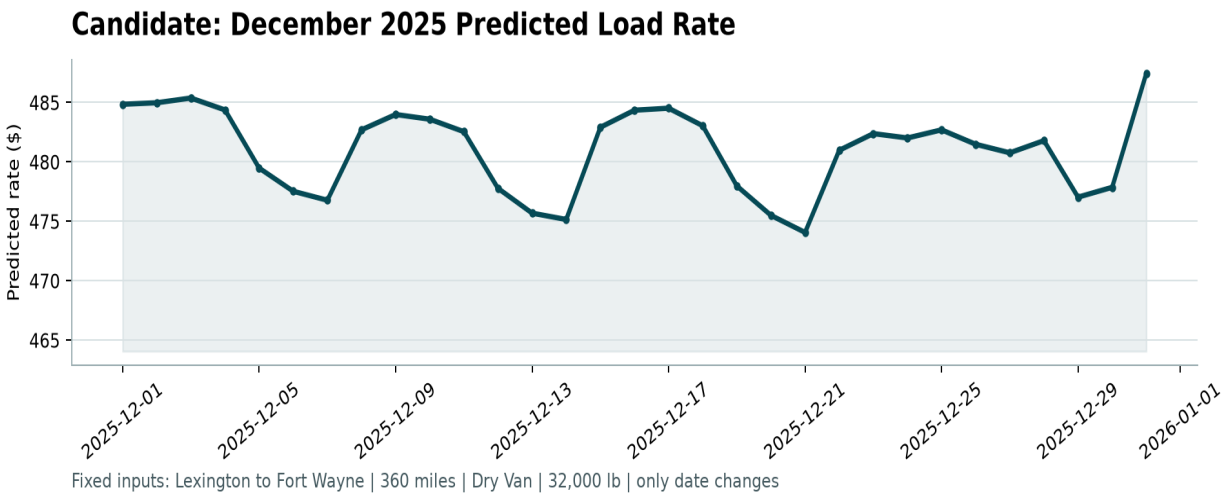
A random split was avoided because the real validation period occurs after the training period. Instead, rows before 1 October 2025 were used for model fitting and October rows were held out as a future-like test set. This produced 43,147 training rows and 4,853 holdout rows. The final selected model was then refit on all 48,000 labeled rows before predicting the actual validation file.

5. Model comparison

Model	Holdout MAE	Decision
HistGradientBoosting	\$193.69	Baseline
CatBoost, direct target	\$153.97	Improved
CatBoost, log target	\$127.09	Selected

CatBoost was selected because it handles categorical variables and route-level effects directly. Training on $\log(1 + \text{posted_rate})$ reduced the influence of extreme rates while predictions were converted back to dollars for submission. No target outliers were deleted automatically.

6. December scenario



The chart shows predicted rates for the fixed Lexington-to-Fort Wayne scenario across December 2025. The scorer validated one positive prediction for every day.

7. Submission artifacts and reproducibility

The complete workflow is documented in [model.ipynb](#). The final load predictions are in [validation_predictions.csv](#), and the December inputs are in [december-chart-inputs.csv](#). The chart is available as [candidate_december.png](#). Dependencies are listed in [requirements.txt](#).

The official scorer confirmed 12,000 valid validation predictions and 31 valid December predictions. Final competition metrics are calculated by Spotter after submission.